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Negative-order Vendi–LCR comparison without a full-rank assumption

Research proof and formalization packet

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Abstract

For every finite positive semidefinite similarity matrix, we prove that the Vendi score bounds the uniform Leinster–Cobbold–Reeve diversity at all negative orders, including the directly defined minus-infinity endpoint. Zero eigenvalues are omitted, following the source's archived implementation. A weighted spectral Jensen inequality gives a stronger bound through the diagonal of the matrix square. The proof requires no full-rank assumption, and the complete matrix-to-score argument is verified in Lean. Known nonnegative-order and full-rank results are credited; classical Jensen and Hölder–McCarthy inequalities are background ingredients. Source-access and prior-art limits are stated explicitly.

1 Definitions and the remaining source case

Let Z be a real symmetric positive semidefinite $N \times N$ matrix, where $N \geq 1$, with

$$Z_{ii} = 1, \qquad 0 \leq Z_{ij} \leq 1.$$

Write $\lambda_1, \ldots, \lambda_N \geq 0$ for its eigenvalues, counted with multiplicity, and put

$$J = \{j : \lambda_j > 0\}, \qquad R_i = \sum_j Z_{ij}, \qquad E_i = (Z^2)_{ii} = \sum_j Z_{ij}^2.$$

Thus $1 \leq E_i \leq R_i$, and J is nonempty. For finite real $q < 1$ define the uniform scores by

$$\mathrm{VS}_q(Z) = \left(\sum_{j \in J} (\lambda_j / N)^q \right)^{1/(1-q)}, \qquad \mathrm{D}_q(Z) = \left(\frac{1}{N} \sum_i (R_i/N)^{q-1} \right)^{1/(1-q)}. \tag{1}$$

The positive-spectrum index set retains eigenvalue multiplicities. For $q < 0$, no negative power is assigned to a zero eigenvalue.

This convention is material. The general displayed spectral formula in Nguyen et al. [1] does not separately explain singular negative powers, but its Appendix C.3 explicitly filters out zero eigenvalues before taking powers, including in searches with negative orders and low-rank Gram matrices. We use that finite support formula. The alternative extended-real convention $0^q = +\infty$ would give a trivial infinite singular comparison; full-rank regularization toward a singular matrix would not prove (1).

Theorem 1 of the latest listed arXiv revision, v4 of March 30, 2026, proves $\mathrm{VS}_q \geq \mathrm{D}_q$ for all $q \geq 0$, and for $q < 0$ when Z is full rank [1]. The public abstract of the May 26, 2026 journal version retains that restriction [2]. The result below supplies the arbitrary-rank negative-order case of the comparison represented by MathDB 371345 and its equivalent power-sum formulation, MathDB 371346. These are formulations of one comparison. No novelty is claimed for the already proved cases.

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2 A spectral energy bound

Theorem 1. *For every matrix Z as above and every real $q < 1$,*

$$\sum_{j\in J} \lambda_j^q \;\geq\; \sum_i E_i^{q-1} \;\geq\; \sum_i R_i^{q-1}. \tag{2}$$

Consequently $\mathrm{VS}_q(Z) \geq \mathrm{D}_q(Z)$ *for every finite real* $q < 1$*; in particular this holds for every* $q < 0$ *at arbitrary rank.*

Proof. Choose a real orthonormal eigenbasis $v_1, \ldots, v_N$, and set $u_{ij} = v_j(i)^2$. Spectral decomposition and column orthonormality give

$$\sum_{j\in J} \lambda_j u_{ij} = 1, \qquad \sum_{j\in J} \lambda_j^2 u_{ij} = E_i, \qquad \sum_i u_{ij} = 1. \tag{3}$$

The omitted zero-eigenvalue terms vanish in the first two identities. For each i, the nonnegative numbers $w_{ij} = \lambda_j u_{ij}$, $j \in J$, therefore sum to one, with weighted mean of the positive eigenvalues equal to E_i.

The function $t \mapsto t^{q-1}$ is convex on $(0, \infty)$ when $q < 1$. Weighted scalar Jensen, followed by summation over i, gives

$$E_i^{q-1} \leq \sum_{j\in J} w_{ij} \lambda_j^{q-1} = \sum_{j\in J} \lambda_j^q u_{ij}, \qquad \sum_i E_i^{q-1} \leq \sum_{j\in J} \lambda_j^q.$$

Since $E_i \leq R_i$ and $q - 1 < 0$, we also have $E_i^{q-1} \geq R_i^{q-1}$, proving (2).

Both normalized interiors in (1) have the same factor:

$$\sum_{j\in J} (\lambda_j/N)^q = N^{-q} \sum_{j\in J} \lambda_j^q,$$
$$\frac{1}{N} \sum_i (R_i/N)^{q-1} = N^{-q} \sum_i R_i^{q-1}.$$

Multiplication by $N^{-q} > 0$ and the positive outer power $1/(1-q)$ preserve the inequality, proving the score comparison. □

The analytic step is classical weighted Jensen. Equivalently, restrict Z to its positive spectral subspace and apply the negative-power Hölder–McCarthy inequality to the unit vector $Z^{1/2} e_i$ (its squared norm is $Z_{ii} = 1$), with exponent $q - 1$. See, for example, the primary treatment in [3, Theorems 1–2]. We do not claim a new general Jensen or operator-power inequality. The point here is the support-sensitive use of that inequality in the remaining Vendi–LCR comparison.

3 The canonical minus-infinity endpoint

Let $m = \min_{j\in J} \lambda_j > 0$ and $r = \min_i R_i > 0$. The canonical endpoint formulas on the same positive support are

$$\mathrm{VS}_{-\infty}(Z) = \frac{N}{m}, \qquad \mathrm{D}_{-\infty}(Z) = \frac{N}{r}. \tag{4}$$

Theorem 2. *For every matrix Z satisfying the stated hypotheses,* $\mathrm{VS}_{-\infty}(Z) \geq \mathrm{D}_{-\infty}(Z)$.

Proof. The weights in (3) express every E_i as a convex combination of positive eigenvalues. Hence $m \leq E_i \leq R_i$ for every i, so $0 < m \leq r$. Taking reciprocals and multiplying by $N > 0$ proves the claim. □

In particular, neither repeated eigenvalues nor rank one requires a separate exception. The statement uses the least *positive* eigenvalue, not the least eigenvalue when zero is present.

4 Formal coverage and prior-art limits

The three modules `VendiJensen`, `VendiMatrix`, and `VendiScores` contain 48 local declarations. A fresh independent replay compiled every declaration with warnings treated as errors using Lean 4.34.1, compiler commit `5045d0056413266e57c625dcd7c365b10e377c52`, and mathlib revision `d13f23b723b8a846827a245b89c10fc7d3f11612` with its pinned dependencies. Axiom inspection found only `propext`, `Classical.choice`, and `Quot.sound`; there are no proof holes, custom axioms, or native computation assumptions.

The certificate starts from an actual finite real matrix satisfying the source hypotheses. It constructs the canonical orthonormal spectral decomposition, removes only zero-weight terms, proves (3), derives the energy and row-sum bounds, and proves the exact normalized comparison. Its endpoints are `source_conjecture_negative` and `source_conjecture_negativeInfinity` in namespace `VendiScores`. Nonempty dimension and diagonal one prove that the positive spectrum is nonempty; its positive minimum and the least row sum are actually selected. No supplied spectral decomposition or invertibility hypothesis remains. The endpoint (4) is certified directly; the packet does not include an analytic theorem about the limit as the order tends to $-\infty$. Known nonnegative-order results are cited background, not additional formally claimed results.

The source and literature screen was conducted on October 26, 2026. We read the complete arXiv v4 primary text and its official public archive, PMC12642762. The later journal PDF required subscription credentials and returned HTTP 401; we read its public primary abstract and do not assert that its full body was checked or is byte-identical to v4. This is a concrete limit on the novelty assessment.

A further prior assertion also deserves acknowledgment. The July 24, 2026 Bioengineering article [4], released in PMC in August, states the VS–LCR comparison without the rank qualification. Its supporting reference 50 is the above journal comparison, and its full public text gives no separate proof of the singular negative-order case. Thus the general comparison was previously asserted; we claim no priority for that assertion. The contribution supported here is a complete proof of the finite support comparison across arbitrary ranks. Targeted primary-source searches found no earlier demonstrated resolution of this case or an explicitly stated complete stronger trace/row-energy bound. This is a bounded prior-art assessment, not an exhaustive priority certificate. The classical tools, the known cases, and the equivalent power-sum formulation are not separate results.

References

[1] P. Nguyen, J. Couch, R. Bansal, A. Morgan, C. Tam, M. Li, R. Arnaout and R. Arnaout, *Which Similarity-Sensitive Entropy (Sentropy)?*, arXiv:2511.03849v4, March 30, 2026; Theorem 1 and Appendix C.3.

[2] P. Nguyen et al., *Comparing frameworks for similarity-aware generalized entropy (sentropy)*, Physical Review E **113** (2026), 055305, doi:10.1103/n2s3-2g85. Public abstract inspected; full journal text not accessed.

[3] S.S. Dragomir, *Jensen type weighted inequalities for functions of selfadjoint and unitary operators*, Italian Journal of Pure and Applied Mathematics **32** (2014), 247–264, primary PDF.

[4] J.D. Couch, R. Arnaout and R. Arnaout, *Beyond Size and Class Balance: Alpha as a New Dataset Quality Metric for Deep Learning*, Bioengineering **13**(8) (2026), 854, doi:10.3390/bioengineering13080854; public full text PMC13510075.

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